CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS
Higher 1

# CHEMISTRY

**8872/01**

**Paper 1 Multiple Choice**

**FRIDAY 16 SEPTEMBER 2011**
**50 Minutes**

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

## READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write and/or shade your name, NRIC / FIN number and HT group on the Multiple Choice Answer Sheet provided.

There are **30 MCQ** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

**Read the instructions on the Answer Sheet very carefully.**

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
Calculators may be used.

## Section A

For each question there are four possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct.

1 In an attempt to establish the formula of an oxide of nitrogen, a known volume of the pure gas, $N_xO_y$ was converted to ammonia and water according to the following equation:

$$N_xO_y \rightarrow x\ NH_3 + y\ H_2O$$

2400 $cm^3$ of the oxide of nitrogen, measured at room temperature and pressure (r.t.p.) produced 7.20 g of water. The ammonia produced was neutralised by 200 $cm^3$ of 1.0 mol $dm^{-3}$ HC*l*. What is the formula of the oxide of nitrogen?

**A** NO **B** $NO_2$ **C** $N_2O$ **D** $N_2O_4$

2 The relative atomic mass of boron, which consists of the isotopes $^{10}B$ and $^{11}B$ is 10.8. What is the percentage $^{11}B$ atoms in the isotopic mixture?

**A** 0.8% **B** 8.0% **C** 20% **D** 80%

3 Nickel-Cadmium rechargeable batteries were once commonly used in handphones and electric shavers. The reaction that occurs when the battery is used is:

$$NiO_2 + Cd + 2\ H_2O \rightarrow Ni(OH)_2 + Cd(OH)_2$$

When the battery goes flat, it can be recharged and the reaction is reversed, restoring the $NiO_2$ and Cd.

What is the change in the oxidation state of nickel when the battery is being used?

**A** from +4 to +2
**B** from +2 to +4
**C** from +2 to +1
**D** from +4 to +1

4 Which of the following electronic configurations represents an element that most likely forms a simple ion with a charge of -3?

**A** $1s^2\ 2s^22p^6\ 3s^23p^1$
**B** $1s^2\ 2s^22p^6\ 3s^23p^3$
**C** $1s^2\ 2s^22p^6\ 3s^23p^63d^3\ 4s^2$
**D** $1s^2\ 2s^22p^6\ 3s^23p^63d^7\ 4s^2$

5 *Use of the Data Booklet is relevant to this question.*

The successive ionisation energies, in kJ $mol^{-1}$, of an element **X** is given below:

870 1800 3000 3600 5800 7000 13200

What is the identity of **X**?

**A** $_{8}O$ **B** $_{33}As$ **C** $_{52}Te$ **D** $_{53}I$

6 In which one of the following pairs do the molecules have similar shapes?

**A** $AlCl_3$ and $PCl_3$

**B** $AlCl_3$ and $BF_3$

**C** $BeCl_2$ and $H_2O$

**D** $BF_3$ and $NH_3$

7 In which one of the following pairs of species is the melting point of the first member lower than that of the second member?

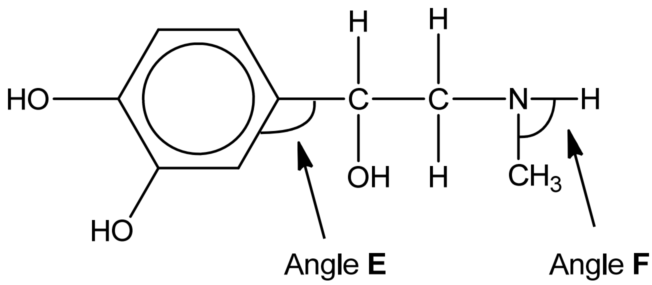
**A** diamond, graphite

**B** $H_2O$, $NH_3$

**C** $Br_2$, $ICl$

**D** pentane, 2,2-dimethylpropane

8 Adrenaline is a hormone which, when secreted directly into the blood stream, acts as a stimulant. It has the following structure:




What are the values of angle **E** and angle **F** in a molecule of adrenaline?

| | angle **E** | angle **F** |
|---|---|---|
| **A** | 120° | 107° |
| **B** | 120° | 90° |
| **C** | 109° | 107° |
| **D** | 109° | 90° |

9 The standard enthalpy changes for two reactions are given by the equations:

$2\,Fe(s) + \frac{3}{2}O_2(g) \rightarrow Fe_2O_3(s)$ $\Delta H^\theta = -822\ kJ\ mol^{-1}$

$C(s) + \frac{1}{2}O_2(g) \rightarrow CO(g)$ $\Delta H^\theta = -110\ kJ\ mol^{-1}$

Using the above information, what is the standard enthalpy change for the following reaction?

$$Fe_2O_3(s) + 3\,C(s) \rightarrow 2\,Fe(s) + 3\,CO(g)$$

**A** $-712\ kJ\ mol^{-1}$

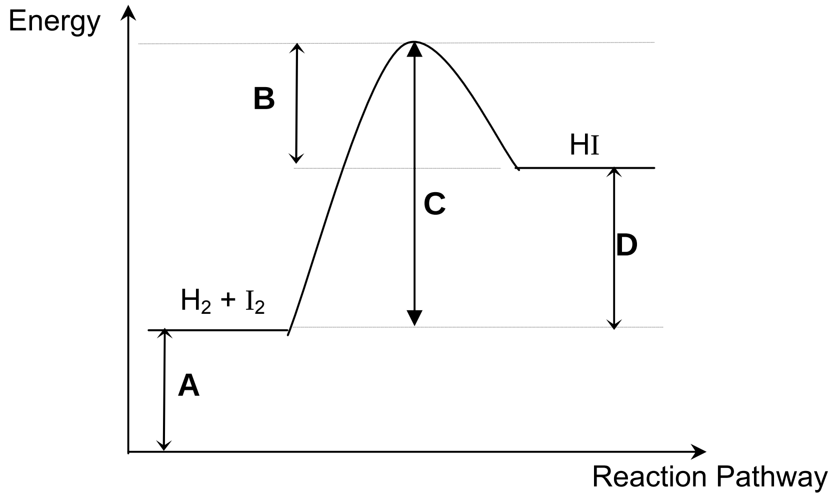
**B** $-492\ kJ\ mol^{-1}$

**C** $+492\ kJ\ mol^{-1}$

**D** $+712\ kJ\ mol^{-1}$

**10** The reaction pathway diagram for the reaction $H_2(g) + I_2 \rightarrow 2HI(g)$ is shown below.

Which correctly labels the activation energy of this reaction?




**11** In the Haber process, ammonia is manufactured according to the following equation:

$$N_2(g) + 3\,H_2(g) \rightleftharpoons 2\,NH_3(g)$$

Which of the following gives the correct condition used in this reaction?

**A** 100°C
**B** 150 atm
**C** Fe catalyst
**D** 1 mol dm$^{-3}$ of $H_2$

**12** How much water must be added to 2.0 dm$^3$ of a solution of pH 2.0 in order to increase its pH to 3.0?

**A** 1.0 dm$^3$ **B** 10 dm$^3$ **C** 18 dm$^3$ **D** 20 dm$^3$

**13** For the equilibrium shown below,

$$H_3O^+(aq) + HPO_4^{2-}(aq) \rightleftharpoons H_2O(l) + H_2PO_4^-(aq)$$

which of the following is a conjugate Bronsted-Lowry acid-base pair?

| | **Base** | **Conjugate Acid** |
|---|---|---|
| **A** | $HPO_4^{2-}$ | $H_3O^+$ |
| **B** | $HPO_4^{2-}$ | $H_2PO_4^-$ |
| **C** | $H_3O^+$ | $H_2O$ |
| **D** | $H_3O^+$ | $H_2PO_4^-$ |

**14** The rate of reaction between a strip of magnesium metal and 50.0 $cm^3$ of 1 mol $dm^{-3}$ dilute hydrochloric acid is determined at 25°C.

In which of the following cases would both sets of conditions bring about an increase in the reaction rate?

**A** Magnesium powder and 100 $cm^3$ of 1 mol $dm^{-3}$ dilute hydrochloric acid.

**B** Magnesium powder and 50.0 $cm^3$ of 0.8 mol $dm^{-3}$ dilute hydrochloric acid.

**C** 100 $cm^3$ of 1.0 mol $dm^{-3}$ dilute hydrochloric acid at 30°C.

**D** 50.0 $cm^3$ of 1.2 mol $dm^{-3}$ dilute hydrochloric acid at 30°C.

**15** Two substances **P** and **Q** react according to the following equation:

$$P + 2\,Q \rightarrow R$$

To determine the order of reaction with respect to each substance, four experiments were performed and the results are shown below:

| **Experiment** | **[P]/mol $dm^{-3}$** | **[Q]/mol $dm^{-3}$** | **Initial rate of formation of R/ mol $dm^{-3}$ $min^{-1}$** |
|---|---|---|---|
| 1 | 0.10 | 0.10 | 0.0010 |
| 2 | 0.40 | 0.10 | 0.0160 |
| 3 | 0.10 | 0.20 | 0.0020 |
| 4 | 0.20 | 0.20 | $x$ |

Which of the following gives the correct value of $x$?

**A** 0.0040 **B** 0.0080 **C** 0.0160 **D** 0.0320

**16** Elements **X**, **Y** and **Z** react with oxygen to form their respective oxides.

When dissolved in water, oxide of **X** gives a strongly alkaline solution while oxide of **Z** gives a strongly acidic solution. Oxide of **Y** is able to react with both acids and bases.

Given that elements **X**, **Y** and **Z** are Period 3 elements, which of the following is **not** true for these three elements?

**A** The atomic radius decreases from **X** to **Y** to **Z**.

**B** The electronegativity increases from **X** to **Y** to **Z**.

**C** The first ionisation energy increases from **X** to **Y** to **Z**.

**D** **X**, **Y** and **Z** could be sodium, magnesium and aluminium respectively.

**17** Which of the following is **not** true about phosphorus?

**A** It has a lower melting point than sulfur.

**B** It can exhibit +3 and +5 oxidation states.

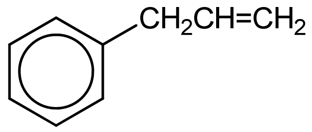
**C** It can form chlorides that hydrolyse in water.

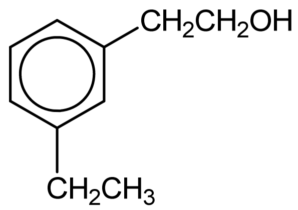
**D** It can form oxides with giant covalent structures.

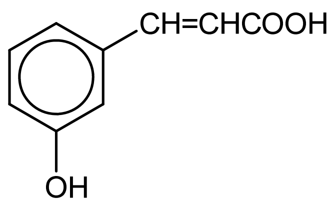
**18** What is the total number of isomers (including stereoisomers) for a non-cyclic organic compound, $C_3H_5Cl$?

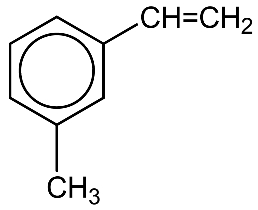
**A** 2 **B** 3 **C** 4 **D** 5

**19** Which one of the following will **not** liberate 2 moles of carbon dioxide when 1 mole of the compound is treated with excess hot acidified potassium manganate(VII)?

**A** 

**B** 

**C** 

**D** 

**20** Which of the following reagents may be used to distinguish between 1-bromobutane and 2-bromobutane?

**A** alkaline aqueous iodine

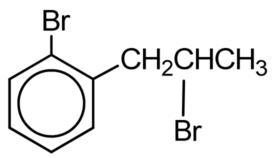
**B** sodium hydroxide

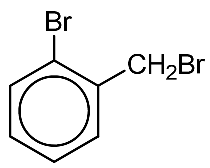
**C** acidified potassium manganate(VII)

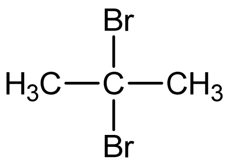
**D** aqueous silver nitrate

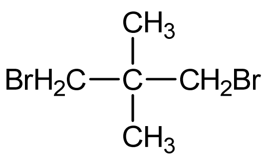
**21** 1 mole of an organic compound **P** undergoes a reaction with alcoholic sodium hydroxide to form 2 moles of HBr.

Which of the following could be **P**?

**A** 

**B** 

**C** 

**D** 

**22** A student was tasked to determine the halogen (either C*l*, Br or I) present in a sample of liquid halogenoalkane. He added 1 $cm^3$ of the sample to 1 $cm^3$ of aqueous NaOH in a test-tube, followed by heating the test-tube in a water bath for 5 minutes. He then added 1 $cm^3$ of dilute HC*l* to neutralise any excess NaOH. Finally, he added 3 drops of aqueous $AgNO_3$ and observed the colour of the precipitate formed.

What was the error in the student's experiment?

**A** using 1 $cm^3$ of aqueous NaOH
**B** heating for 5 minutes
**C** using 1 $cm^3$ of dilute HC*l*
**D** using 3 drops of aqueous $AgNO_3$

**23** Which of the following syntheses will produce compound **Y**, 2-methylpropanoic acid?

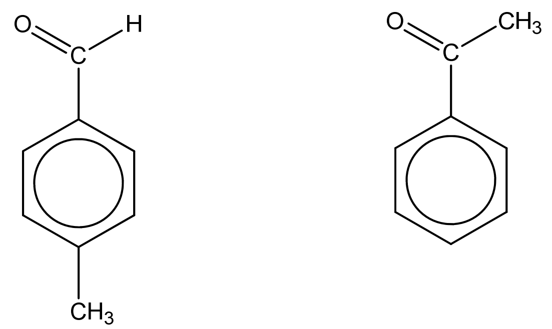
**A** $CH_3CH(OH)CH_3 \xrightarrow[\text{heat}]{Cr_2O_7^{2-}/H^+} \mathbf{X} \xrightarrow[\text{cold}]{HCN/OH^-} \mathbf{Y}$

**B** $CH_3CH(Br)CH_3 \xrightarrow[\text{heat}]{KCN(alc.)} \mathbf{X} \xrightarrow[\text{heat}]{H_2O/H^+} \mathbf{Y}$

**C** $(CH_3)_2C{=}CH_2 \xrightarrow[\text{cold}]{MnO_4^-/H^+} \mathbf{X} \xrightarrow[\text{heat}]{Cr_2O_7^{2-}/H^+} \mathbf{Y}$

**D** $(CH_3)_2CHCH_2Br \xrightarrow[\text{heat}]{KCN(alc.)} \mathbf{X} \xrightarrow[\text{heat}]{H_2O/H^+} \mathbf{Y}$

**24** Which of the following chemical tests will give different observations for the following two compounds?




**A** Warm with 2,4–dinitrophenylhydrazine
**B** Warm with Fehling's solution
**C** Warm with Tollen's reagent
**D** Warm with aqueous bromine

**25** Consider the following four compounds:

1. $CH_3CH_2CH_2OH$
2. $CH_3CH_2CO_2H$
3. $CH_3\underset{\underset{\displaystyle F}{|}}{C}HCO_2H$
4. $CH_3\underset{\underset{\displaystyle Cl}{|}}{C}HCO_2H$

What is the order of increasing acidity of the compounds (weakest to strongest)?

**A** 1 → 2 → 4 → 3

**B** 1 → 2 → 3 → 4

**C** 3 → 4 → 2 → 1

**D** 3 → 4 → 1 → 2

For questions **26** to **30**, one or more of the three numbered statements **1** to **3** may be correct. The responses **A** to **D** should be selected on the basis of

| A | B | C | D |
|---|---|---|---|
| **1**, **2** and **3** are correct | **1** and **2** only are correct | **2** and **3** only are correct | **1** only is correct |

No other combination of statements is used as a correct response.

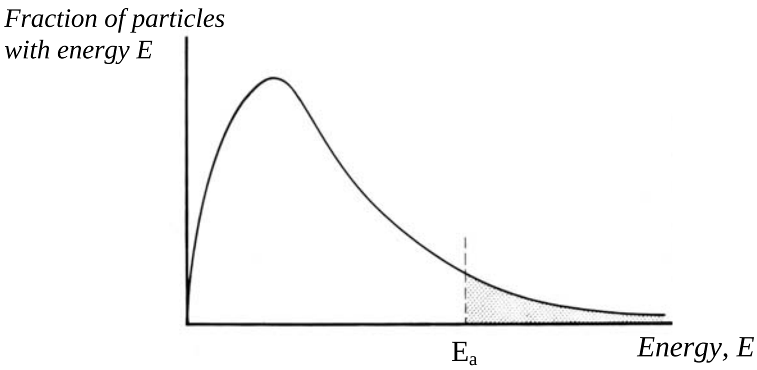
**26** Which of the following substances conduct electricity due to delocalised electrons?

**1** graphite

**2** benzene

**3** molten sodium chloride

**27** The diagram below shows the Boltzmann Distribution of molecular energies at a particular temperature.




When the temperature is increased, which of the following occurs?

**1** The activation energy is lowered.

**2** The graph becomes broader.

**3** The peak shifts to the right.

**28** Germanium is directly below silicon in the Periodic Table.

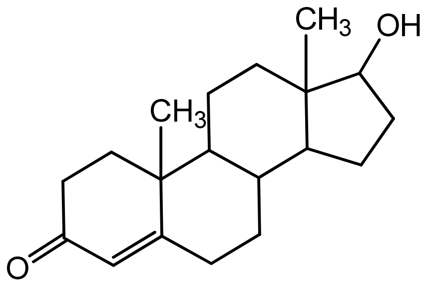
Which statements about germanium are **true**?

**1** It exists as a solid at room temperature.

**2** It has a smaller ionic radius than silicon.

**3** It does not conduct electricity at all temperatures.

**29** Testosterone is one of the most important male sex hormones and its structure is given below.




Testosterone

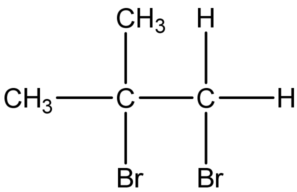
Which of the following statements are true?

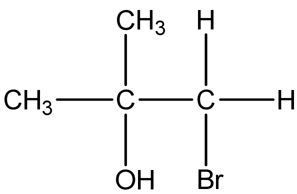
**1** It gives an orange precipitate with 2,4-dinitrophenylhydrazine.

**2** It decolourises orange aqueous $Br_2$.

**3** It reacts with $SOCl_2$ to give steamy fumes of $HCl$.

**30** Which of the following are possible products when 2-methylpropene is reacted with aqueous bromine?

**1** 

**2** 

**3**